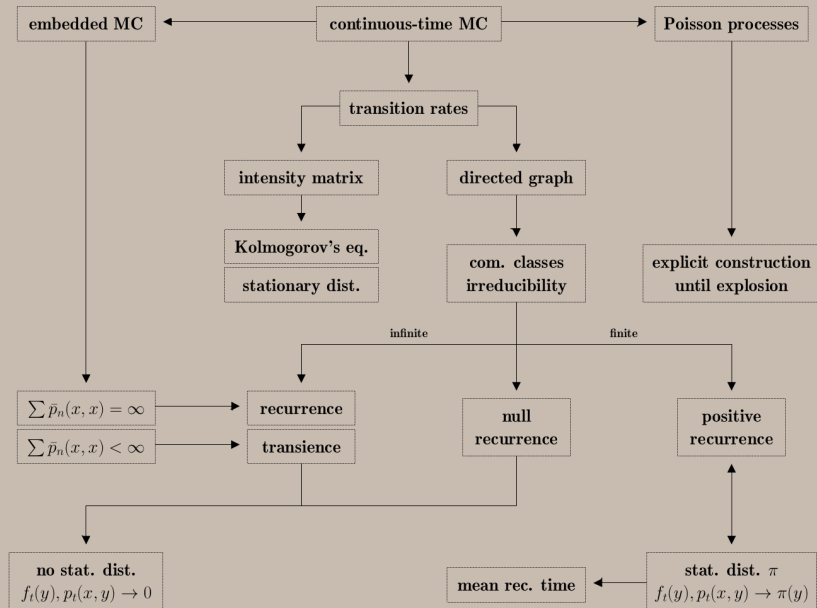
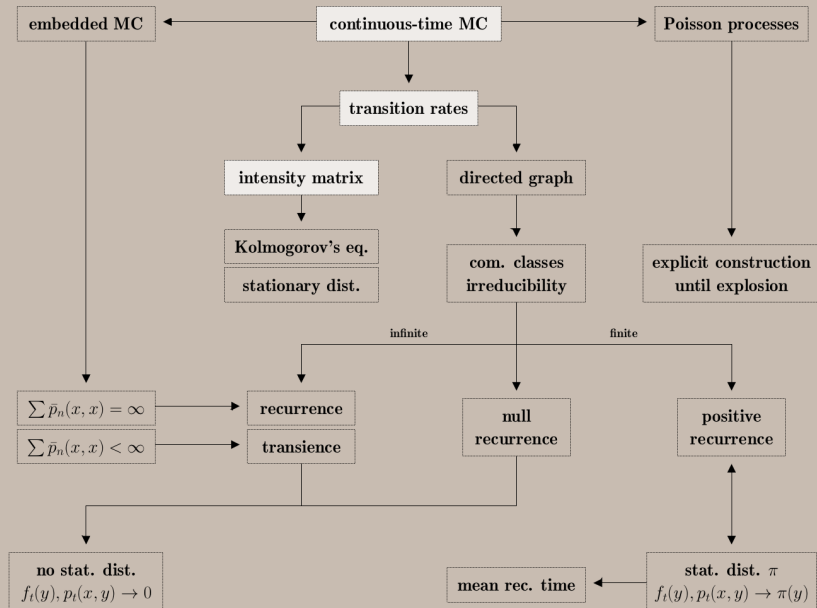




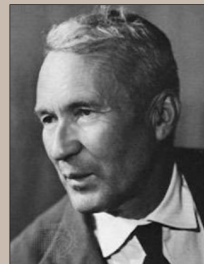
Andrey Markov



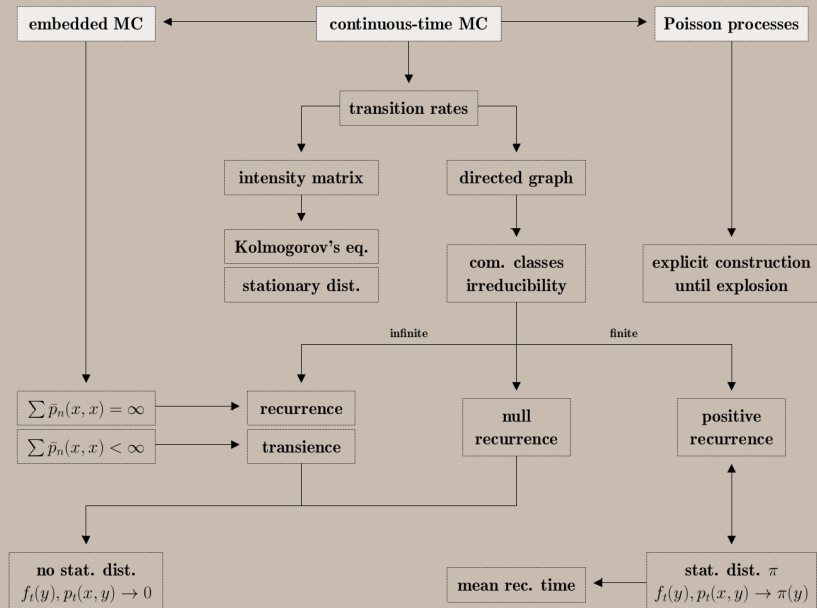
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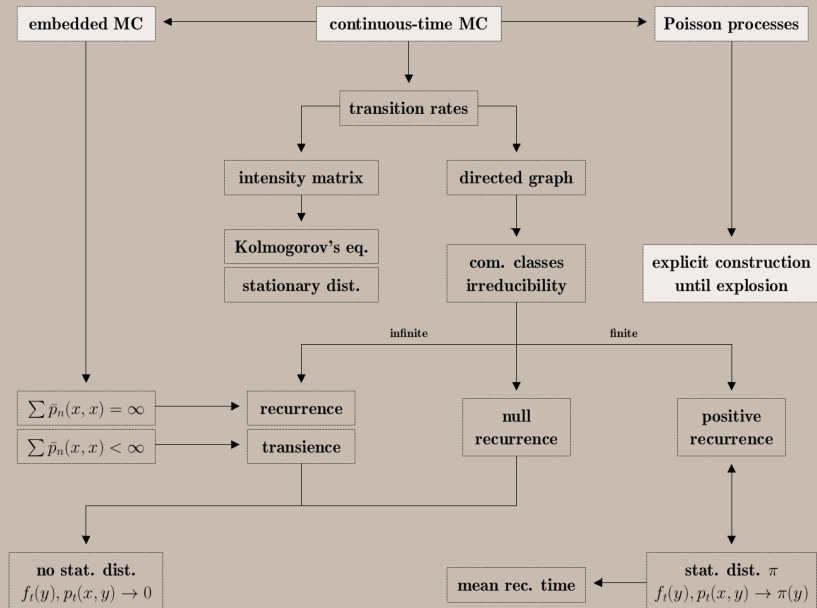
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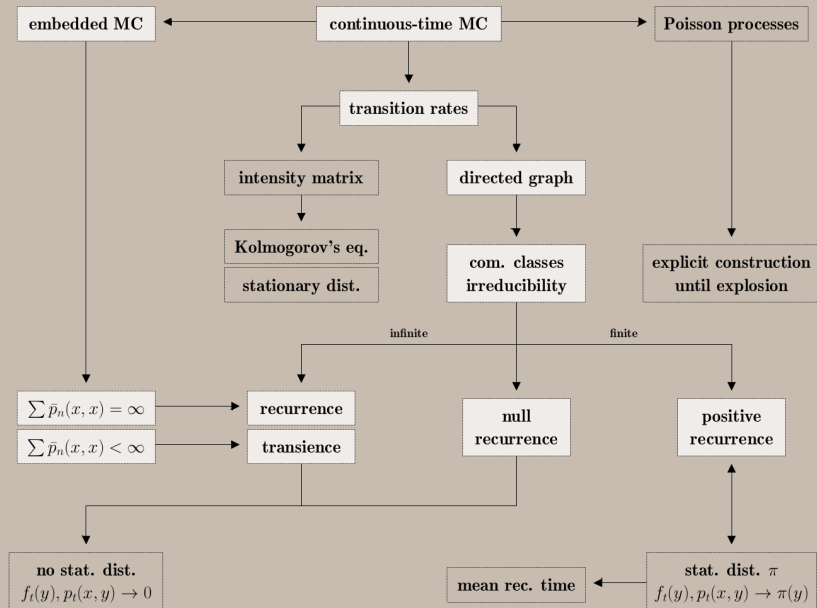
Siméon Poisson



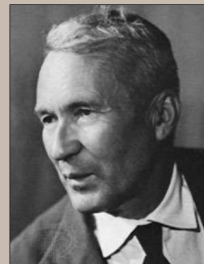
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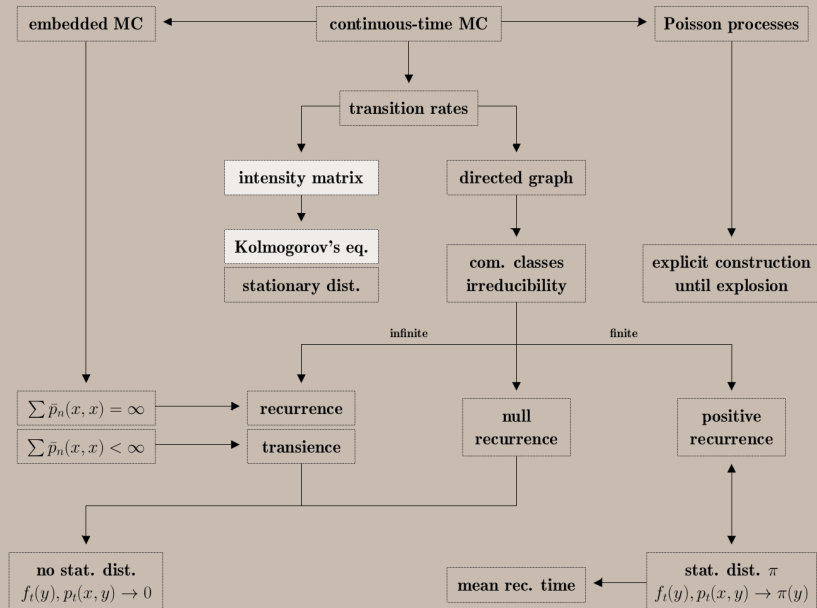
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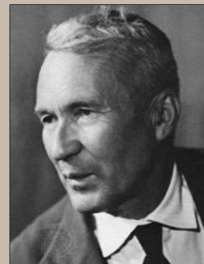
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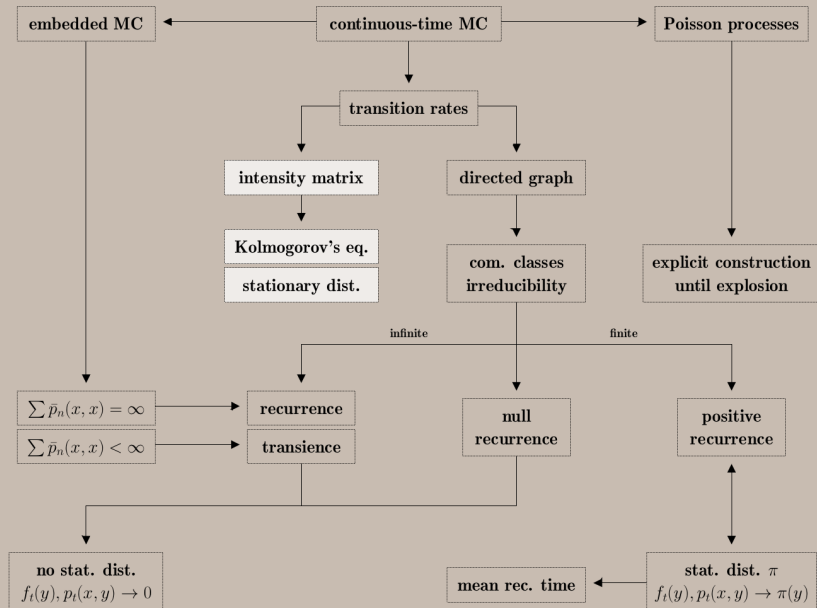
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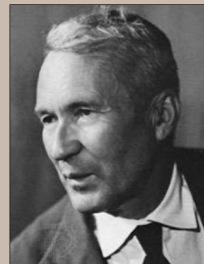
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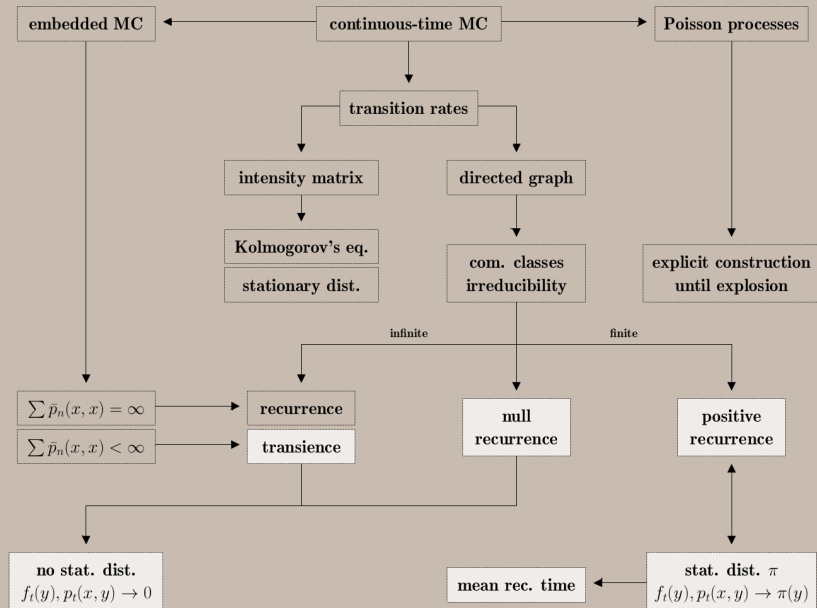
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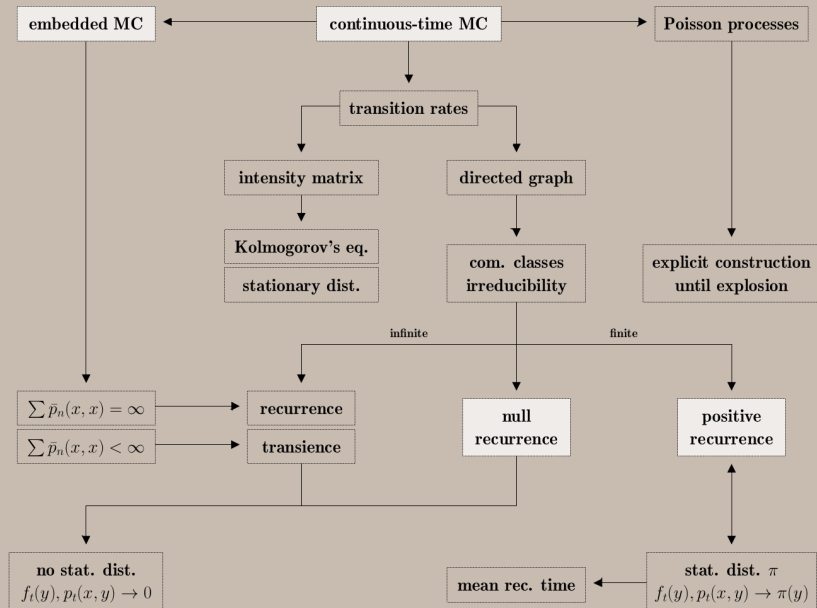
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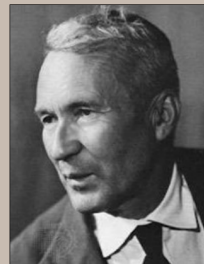
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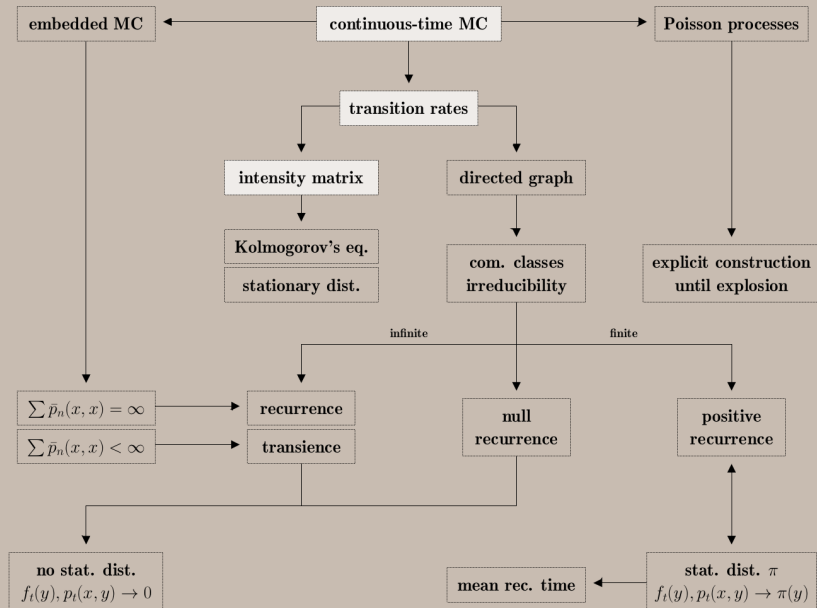
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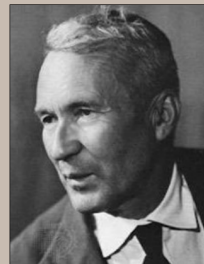
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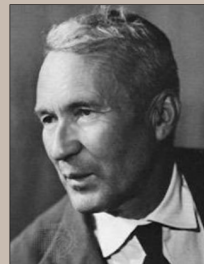
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Definition. The process X_t is a **continuous-time Markov chain** with state space S finite or countable whenever

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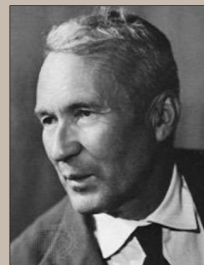
$$P(X_s \in B \mid \mathcal{F}_t) = P(X_s \in B \mid X_t) \quad \text{for all } s > t \text{ and } B \subset S.$$

No successor on $\mathbb{R} \implies$ define full collection of **transition probabilities**

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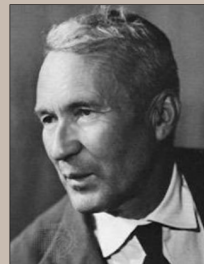
$$p_t(x, y) = P(X_t = y \mid X_0 = x) \quad \text{for all } t > 0 \text{ and } x, y \in S.$$

Solutions to an ODE involving the **transition rates**

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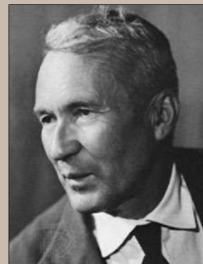
$$c(x, y) = \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} p_\epsilon(x, y) \quad \text{for all } x \neq y.$$

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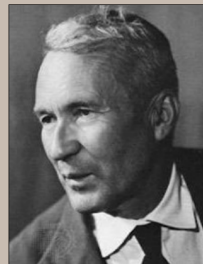
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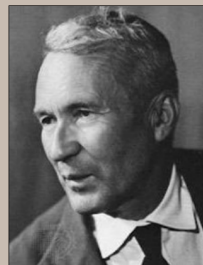
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Definition. **Intensity matrix**

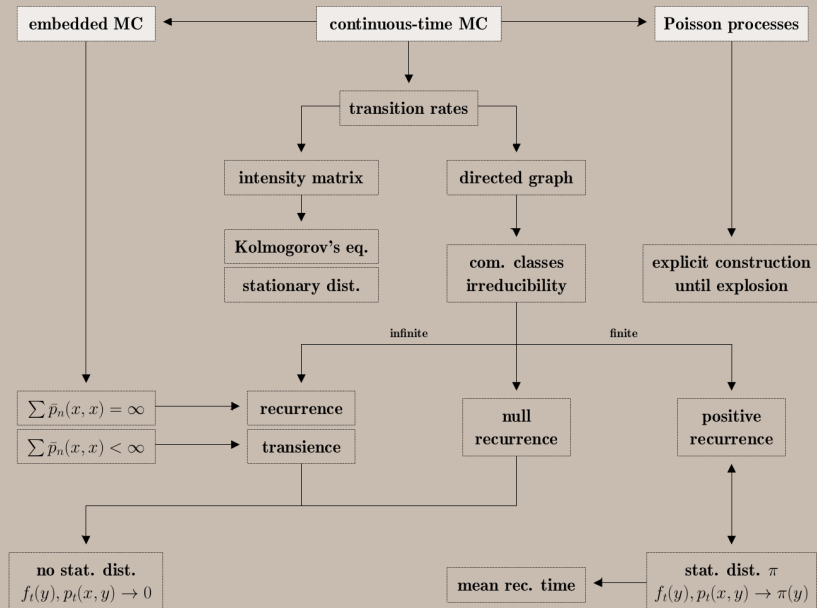
$$Q = (q(x, y))_{x, y \in S} \quad \text{where} \quad q(x, y) = \begin{cases} c(x, y) & \text{for } x \neq y \\ -\sum_{z \neq x} c(x, z) & \text{for } x = y. \end{cases}$$



Andrey Markov



Andrey Kolmogorov



Andrey Markov



Siméon Poisson

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sequence of states visited
described by embedded Markov
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$$\text{Superposition} \implies \begin{cases} \text{time until next jump } \tau = \min_{z \neq x} \tau_z = \text{Exponential}(\sum_{z \neq x} c(x, z)), \\ \text{jump to state } y \text{ with probability } P(\tau = \tau_y) = c(x, y) / \sum_{z \neq x} c(x, z). \end{cases}$$

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Proof (for $t = 0$). For $\epsilon > 0$ small, define

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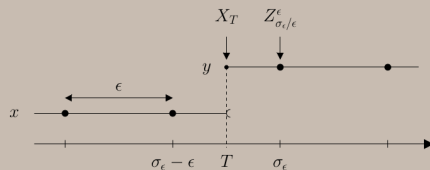
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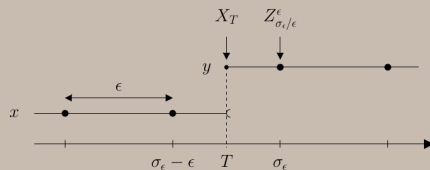
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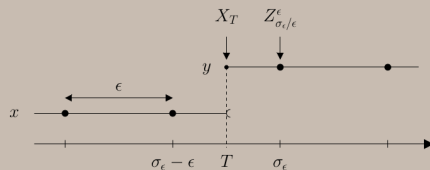
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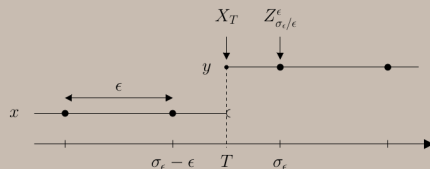
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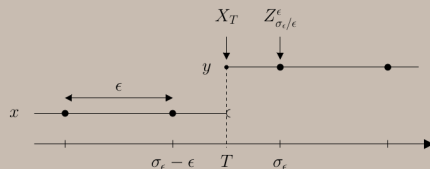
$$\text{Intensity matrix} \implies \begin{cases} p_\epsilon(x, x) = 1 - \sum_{z \neq x} p_\epsilon(x, z) = 1 + \epsilon q(x, x) + o(\epsilon) \\ p_\epsilon(x, y) = \epsilon q(x, y) + o(\epsilon) \end{cases}$$

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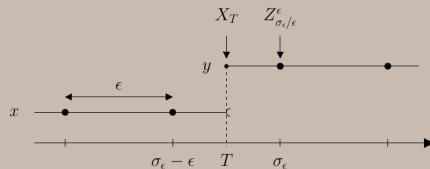


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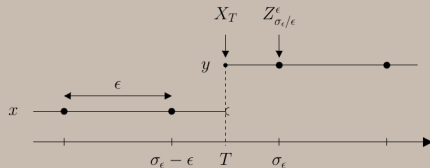
$$\text{Sum (2) over } y \neq x \implies P_x(\sigma_\epsilon = \epsilon n) \stackrel{(2)}{=} \sum_{y \neq x} (p_\epsilon(x, x))^{n-1} p_\epsilon(x, y) = (p_\epsilon(x, x))^{n-1} (1 - p_\epsilon(x, x)).$$

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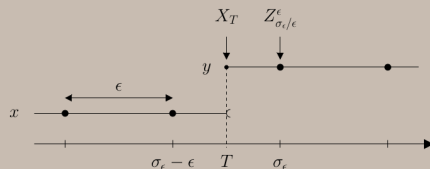
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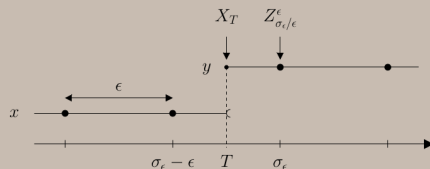
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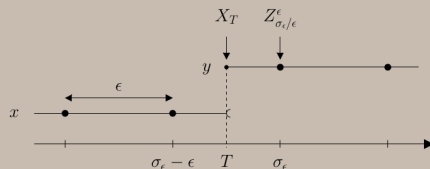
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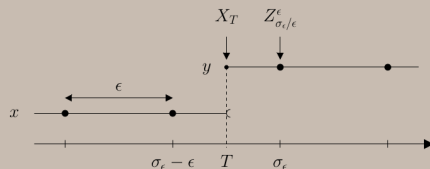
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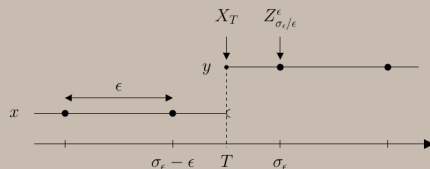
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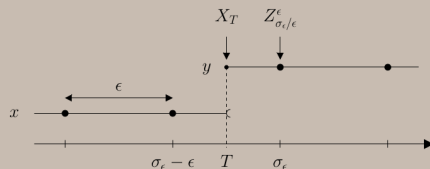
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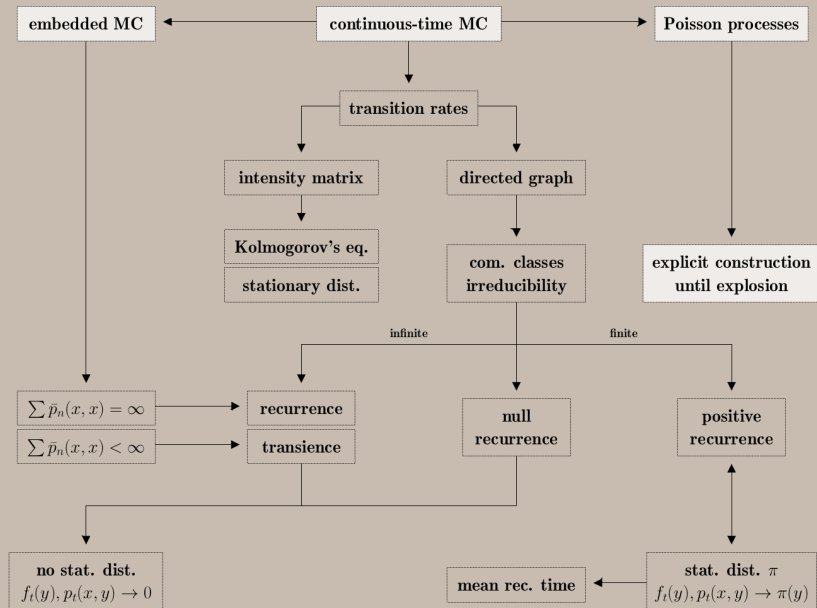


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Andrey Markov



Siméon Poisson

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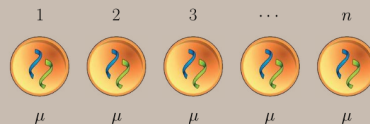
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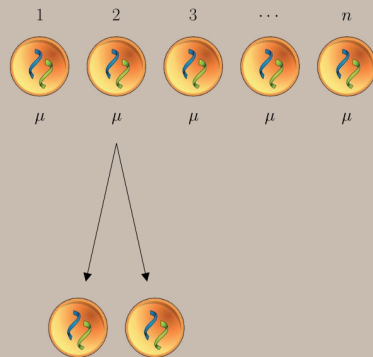
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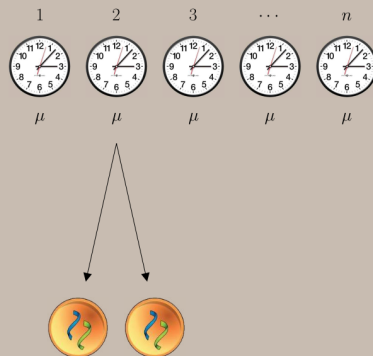
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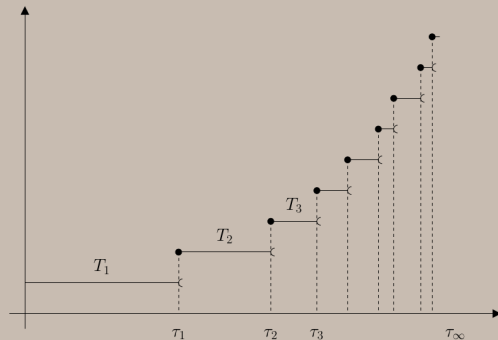
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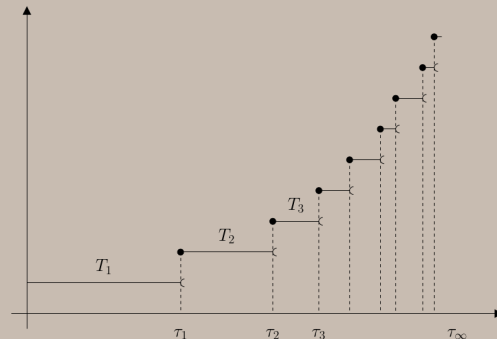
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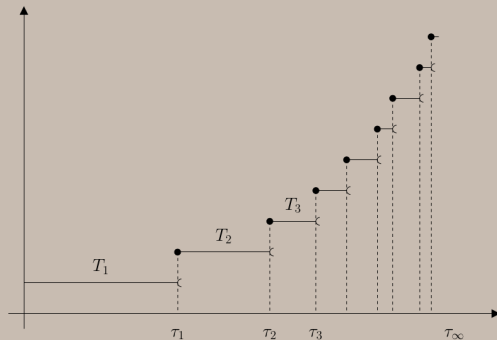
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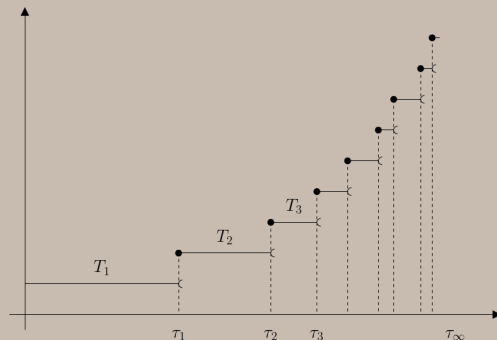
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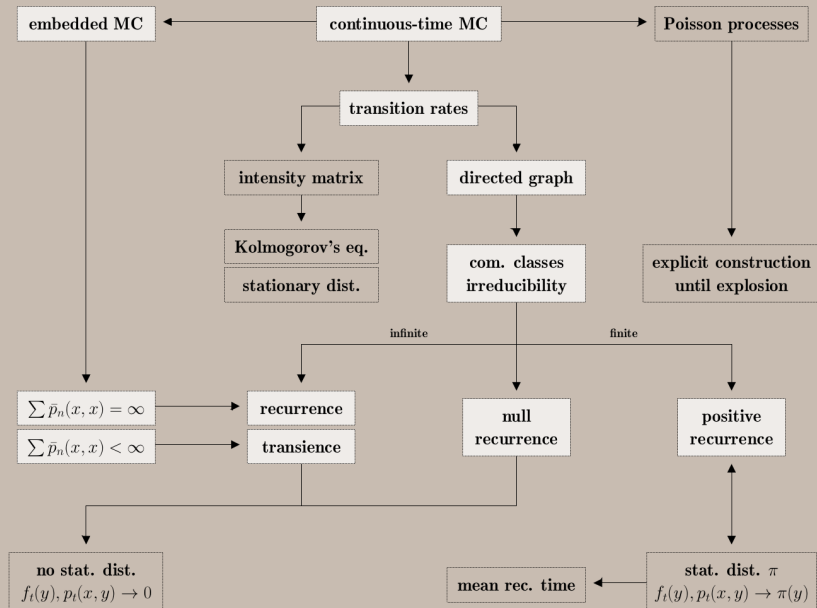
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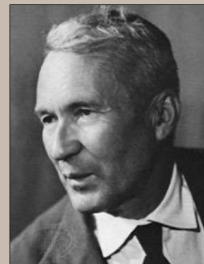
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$\Rightarrow \tau_\infty < \infty$ and so explosion.





Andrey Markov



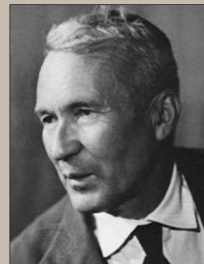
Andrey Kolmogorov

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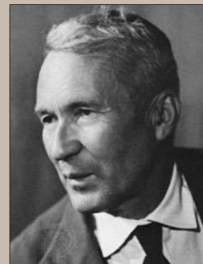
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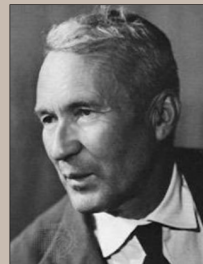
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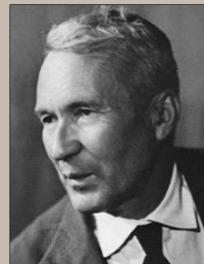
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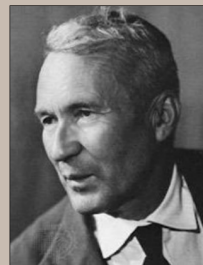
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$$c(x, y) > 0 \quad \Longleftrightarrow \quad \bar{p}(x, y) = c(x, y) / \sum_{z \neq x} c(x, z) > 0$$



Andrey Markov



Andrey Kolmogorov

Definition. States x, y **communicate** ($x \leftrightarrow y$) if

$$p_s(x, y) > 0 \quad \text{and} \quad p_t(y, x) > 0 \quad \text{for some} \quad s, t \geq 0.$$

Definition. The process is **irreducible** if only one communication class.

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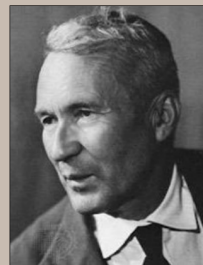
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$\implies X_t$ and Y_n have the same directed graph/**communication classes**.



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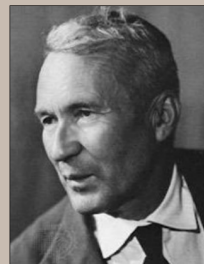
positive recurrent when $P_y(T_y < \infty) = 1$ and $E_y(T_y) < \infty$

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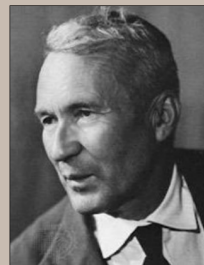
null recurrent when $P_y(T_y < \infty) = 1$ and $E_y(T_y) = \infty$

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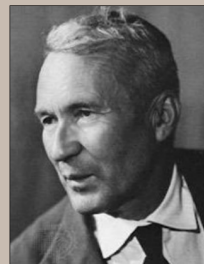
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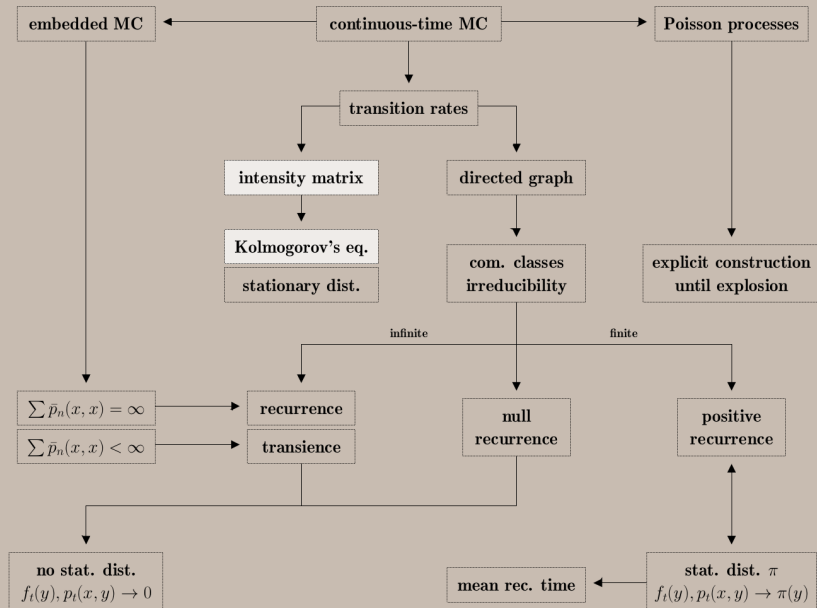
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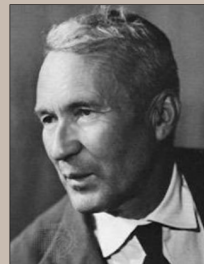
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Chapman-Kolmogorov's equations: condition on $\{X_t = z\}$, $z \in S \implies$

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$$\begin{aligned} p_{t+\epsilon}(x, y) &\stackrel{\text{CKE}}{=} \sum_{z \in S} p_{\epsilon}(x, z) p_t(z, y) \\ &= \sum_{z \neq x} p_{\epsilon}(x, z) p_t(z, y) + p_{\epsilon}(x, x) p_t(x, y) \end{aligned}$$

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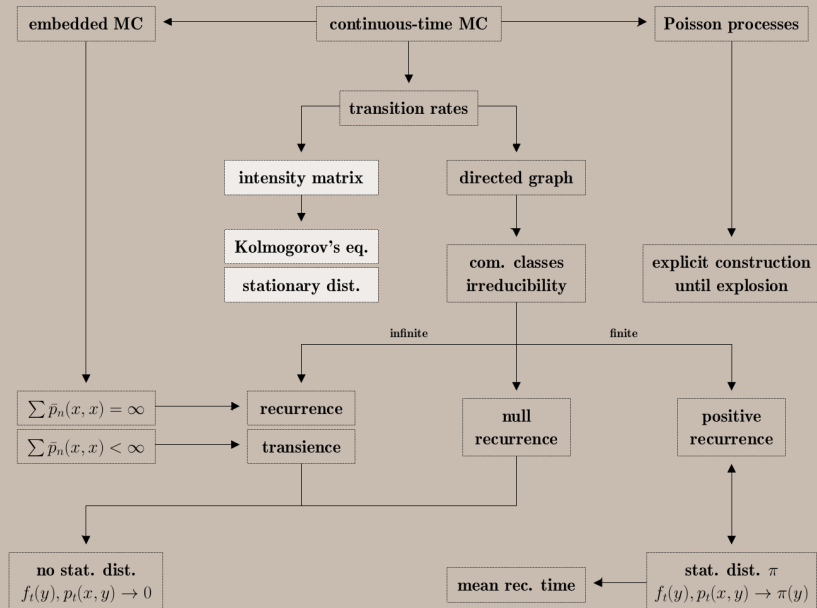
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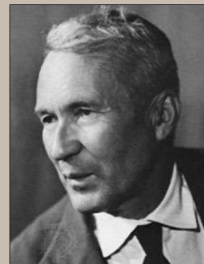
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Andrey Markov



Andrey Kolmogorov

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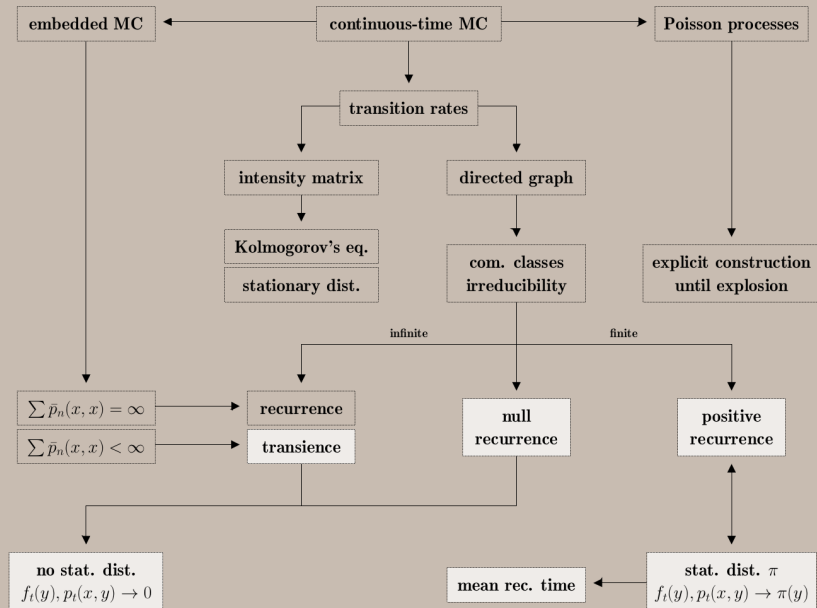
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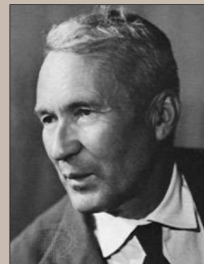
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Reversible distribution: $\pi(x) q(x, y) = \pi(y) q(y, x)$ for all $x \neq y$ (detailed balance condition).

$$(\pi Q)(x) = \sum_{y \in S} \pi(y) q(y, x) \stackrel{\text{DBC}}{=} \sum_{y \in S} \pi(x) q(x, y) = 0 \implies \pi \text{ is stationary.}$$



Andrey Markov



Andrey Kolmogorov

Theorem. Assume **irreducibility** and define $f_t(y) = \frac{1}{t} \int_0^t \mathbf{1}\{X_s = y\} ds$. Then,

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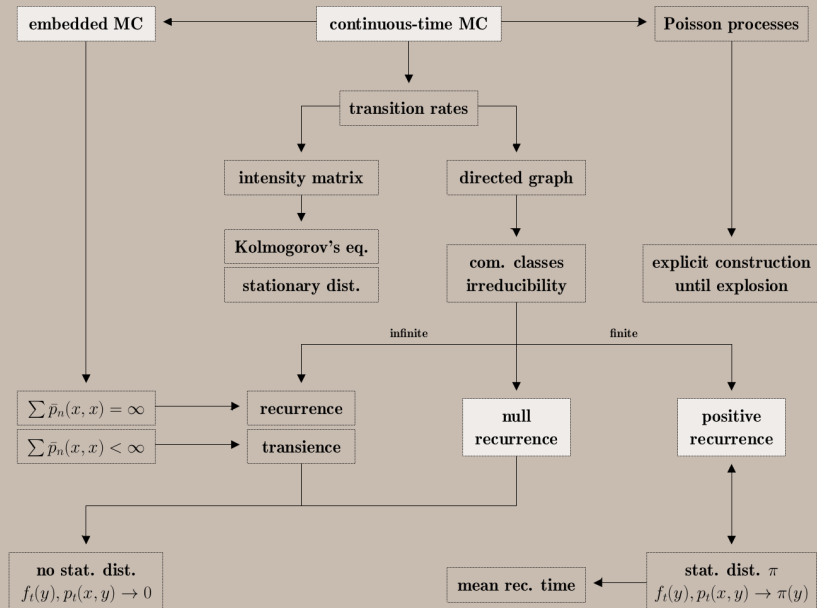
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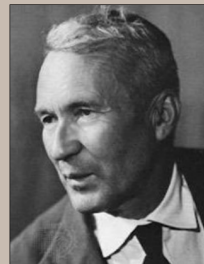
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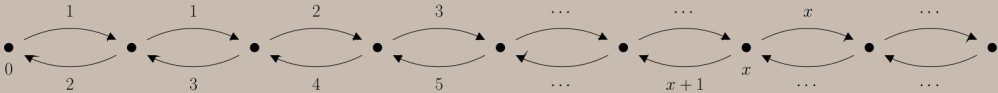
Use results from **discrete-time Markov chains** to transport properties of Z_n^ϵ to the process X_t .

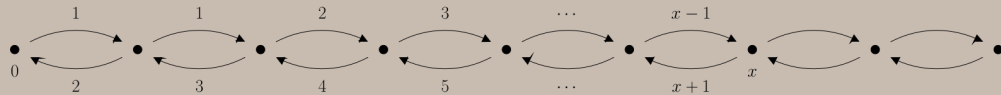


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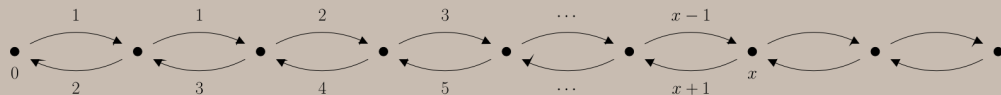
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Continuous-time Markov chain X_t

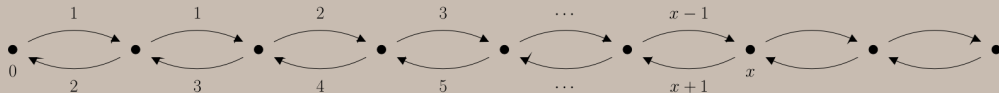
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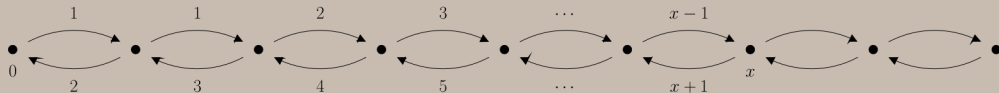
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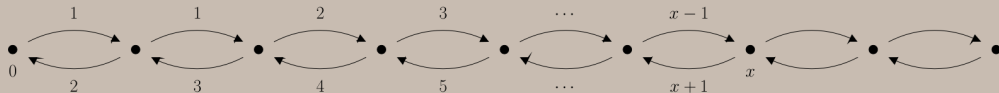


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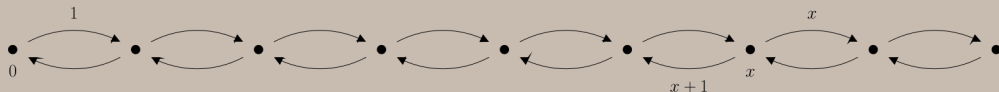


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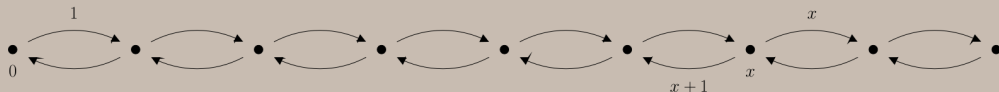
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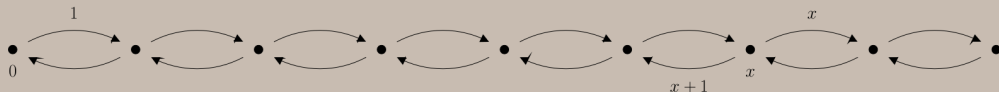
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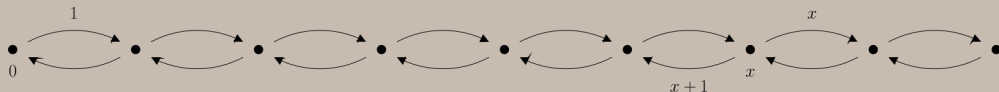
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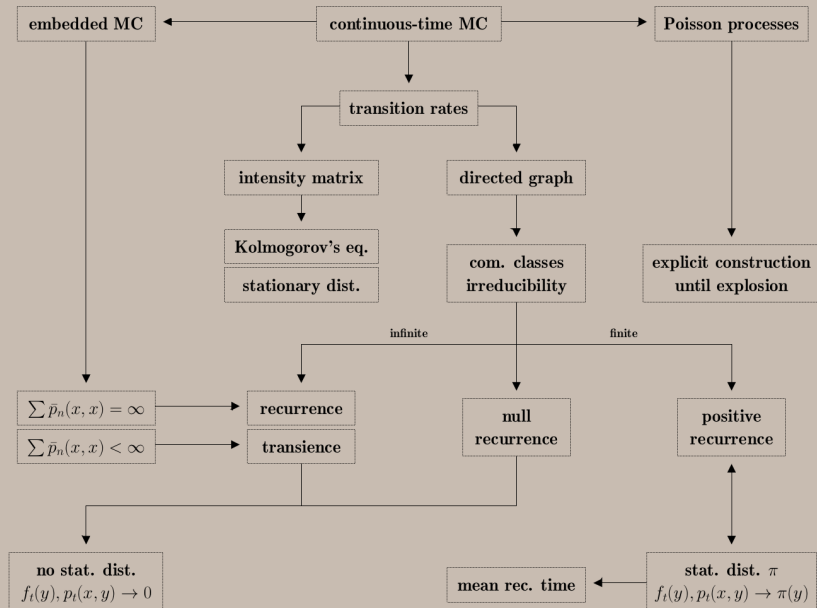
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